

ARNAV VERMA

+1(551) 359-3439 • av4445@nyu.edu

[Linkedin.com/in/namesarnav](https://www.linkedin.com/in/namesarnav) • [Github.com/namesarnav](https://github.com/namesarnav) • namesarnav.com



Education

New York University, Courant Institute of Mathematical Sciences – New York, NY

Expected 05/2028

Master of Science, Computer Science

Coursework: **Advanced Algorithms**, **Machine Learning**, **Deep Reinforcement Learning**

University of North Texas - Denton, TX

01/2023-05/2026

Bachelor of Science, Computer Science, Minor Mathematics; **GPA 3.70**

Coursework: **Data Structures and Algorithms**, **Operating Systems**, **Natural Language Processing**, **Database Systems**

Awards: **Undergraduate Research Fellowship**, **President's List**, **Winner HackUNT**, **HackSMU**, **HackUTD**

Skills

Languages	C/C++, Python, Go, SQL, Bash, HTML, CSS, JavaScript, TypeScript
ML/AI	PyTorch, TensorFlow, HuggingFace Transformers, LangChain, Scikit
Data	PostgreSQL, SQLite, Redis, DynamoDB, FAISS, Pinecone
Backend	FastAPI, Express, Prisma, Flask
Tools	Docker, Kubernetes, Git, GitHub Actions, Weights and Biases, MLFlow
Certifications	CNCF Certified Kubernetes Application Developer (CKAD), Cred ID: LF-5ridapwt2z

Publications

- Iqbal, F., Das, S., Dutt, R., **Verma, A.**, Ray Choudhury, S. "Evaluating Multi-Dimensional Generalization of Large Language Models in Temporal Extraction Tasks." **Accepted, AACL-IJCNLP Findings 2026.**
- Verma, A.**, Iqbal, F., Dutt, R., Ray Choudhury, S. "Do Large Language Models Reason Causally, or Just Pattern Match? A Multi-Strategy Evaluation" **Under review**

Experience

Undergraduate Research Fellow – University of North Texas

05/2025-05/2026

- Evaluated **42 model configurations** (LLaMA, Qwen, Mistral) across dense and MoE architectures on temporal and causal reasoning tasks, designing a Hungarian algorithm based span matching metric for rigorous F1 evaluation across **7+ generalization datasets**
- Led systematic analysis of reasoning strategies (inductive, deductive, abductive) across **7 generalization dimensions**, identifying that inductive prompting improved cross-dimensional consistency and outperformed chain of thought style combinations
- Developed adversarial and domain shift evaluation datasets (TextFooler, DeepWordBug, clinical domain transfer) that exposed up to **60%+** performance drops under distribution shift, **fine-tuning RoBERTa/T5/GPT-2** baselines using HuggingFace Transformers to benchmark against prompted LLMs

Projects

SEMANTIC VIDEO SEARCH ENGINE — PYTHON, FAISS, CLIP, SQLite

[Code](#) | [Demo](#)

- Engineered a CLIP based video search engine over **205 videos and 4,415 frames**, returning ranked, playable timestamps from plain language queries
- Made a query path encoding text with CLIP, ranking **4,415 normalized vectors** by FAISS inner product, then joining SQLite metadata
- Produced scene cut sampling on **Bhattacharyya distance** in HSV histograms, threshold at **0.35** inside a measured **0.20–0.65** plateau
- Proved scene aware sampling raises **Recall@5** from **85.7% to 100%** on a purpose built cut dense benchmark

MARIGOLD — PYTHON, TYPESCRIPT, GOOGLE GEMINI, POSTGRES, FLASK

[Code](#) | [Demo](#)

- Built and deployed a PDF to Quiz study platform serving **500 users**, using Flask, Postgres, Redis and React, generating 15 flashcards per upload through Gemini calls
- Deployed a SAKT knowledge tracing model at **0.756 held out AUC across 277,720** interactions from **3,022 students**, reproducing the training run end to end
- Eliminated a **100% error rate** at **90 concurrent users**, raising authenticated throughput from **0.9 to 604 requests per second**
- Proved correctness across 357 tests, running 245 against Postgres in CI as SQLite silently ignores the foreign keys production enforces

POLISH – TYPESCRIPT, EXPRESS, REACT, POSTGRES, API DEVELOPMENT

[Code](#) | [Demo](#)

- Developed a **23 endpoint** Express and Prisma REST API backing an AI resume editor across **5 PostgreSQL models**
- Engineered **86 unit tests** across 8 suites, reaching **100% statement** coverage on all five service modules and auth middleware
- Exposed a positional diff that turned a 1 line insertion into 101 changed rows with 0 unchanged lines
- Validated LLM output handling with 17 tests covering markdown fenced JSON, malformed responses, and empty array fallback
- Quantified bcrypt cost factor **12 at 222 ms** per hash, nearly **4x the 58 ms** of the common default